CHIRAG BATRA

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PROFESSIONAL SUMMARY

Computer Science (Data Science) undergraduate (batch of 2027) who builds ML systems end to end in Python: data pipelines, model training, explainability, and REST API deployment. Two internships, a live-deployed ML web app, and Hireloop, an LLM job copilot with retrieval, agents, and evals, in progress. Seeking an AI Engineer role.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, JavaScript
Machine Learning and AI: Scikit-learn, XGBoost, Random Forest, Logistic Regression, Uplift Modeling (CausalML), SHAP, Feature Engineering, Model Evaluation, LLM Applications, RAG, Agentic AI, LangGraph
Data and Pipelines: Pandas, NumPy, ETL Pipelines, Data Preprocessing, Exploratory Data Analysis, SQLAlchemy
Deployment and MLOps: REST APIs, FastAPI, Flask, MLflow, Docker, GitHub Actions, Render, Git, GitHub
Databases and Tools: SQL, SQLAlchemy, PostgreSQL (Supabase), Streamlit, React, Jupyter Notebook
Computer Science Fundamentals: Data Structures and Algorithms, OOP, Database Management Systems

WORK EXPERIENCE

Project Intern, Data Science
IBM | Remote | Jun 2026 - Jul 2026
- Built EventLens AI, a full-stack financial risk platform in Python computing Value-at-Risk and Monte Carlo portfolio projections.
- Trained a Random Forest (Scikit-learn pipeline) on 250+ event-days of GDELT news data to predict abnormal returns from news sentiment and volatility features.
- Engineered ETL pipelines joining news events with live market prices in a SQLAlchemy database, served through FastAPI REST APIs to a React reporting dashboard.

Research Intern, Data Analysis
ISVR Aerospace Projects | Dehradun, India | Jun 2025 - Jul 2025
- Engineered Python pipelines to remove drift and noise from four sensor telemetry streams (GPS, IMU, pressure, temperature).
- Automated quality-control checks that detected altitude-control anomalies in time-series sensor data.
- Compared five operational scenarios and quantified performance trade-offs for technical stakeholders.

PROJECTS

Hireloop: AI Job Copilot (Ongoing) | github.com/thechiragbatra/hireloop
Python, FastAPI, LangGraph, React, TypeScript, PostgreSQL, Docker, GitHub Actions
- Building a full-stack AI job copilot that parses a resume into a structured profile, ranks jobs from public ATS APIs (Greenhouse, Lever), and tailors the resume per role using only facts from the user's profile.
- Designing a hybrid retrieval pipeline (BM25 + embeddings + reranking) and a LangGraph workflow with human-in-the-loop approval and Pydantic-validated structured LLM outputs.
- Adding a faithfulness verifier that flags unsupported claims, Supabase Auth with row-level security, rate limiting, and GitHub Actions CI; backend targeted for Render and frontend for GitHub Pages.

Proactive Academic Intervention System | github.com/thechiragbatra/proactive-academic-intervention-system
Python, Scikit-learn, Flask, Render | Team of 3
- Developed a full-stack ML web application (Flask, deployed live on Render) that scores 5,000 student records with a hybrid risk model (70% ML, 30% rules), achieving 0.887 ROC-AUC, 0.764 F1, and 79.9% recall on at-risk students.
- Engineered four derived features and implemented six DSA modules (max-heap priority queue, sliding-window attendance anomaly detector, hash map, graph BFS, greedy planner, mergesort), covered by 11 pytest tests.

Customer Churn Prediction and Retention Uplift Modeling | github.com/thechiragbatra/customer-churn-mlops
Python, XGBoost, CausalML, SHAP, MLflow, FastAPI, Streamlit
- Built a churn model on 7,032 Telco customers using XGBoost with SMOTE for imbalance, achieving 0.82 AUC-ROC.
- Applied T-learner uplift modeling (CausalML) to segment customers into Persuadable, Sure Thing, Lost Cause, and Sleeping Dog groups, cutting simulated retention-campaign spend by 56 percent on synthetic treatment data.
- Built a Streamlit dashboard with SHAP explainability and a FastAPI prediction API; tracked experiments in MLflow.

EDUCATION

Bachelor of Technology in Computer Science (Data Science)
UPES Dehradun | Dehradun, India | 2023 - 2027

CERTIFICATIONS

- Agentic AI Certified Foundations Associate, Oracle University (Oct 2026)
- Data Science Tools, IBM SkillsBuild
- Data Analytics Virtual Experience Program, Deloitte Australia (Forage)
- Data Structures and Algorithms using Python, Tutedude